PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd .under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for

B.Voc. (Horticulture Sciences)

As per NEP 2020

2026-2027

Course Outline for B.Voc. (Horticulture Sciences)
As per NEP 2020

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 101	Core Discipline	Weed Control	2	2	4	40	10	40	10	100
HSC 102	Core Discipline	Nursery Production	2	2	4	40	10	40	10	100
HSC 103	Core Discipline	Irrigation and Water Management	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 201	Core Discipline	Propagation Techniques	2	2	4	40	10	40	10	100
HSC 202	Core Discipline	Elements of Vegetable Growing	2	2	4	40	10	40	10	100
HSC 203	Core Discipline	Horticulture (Fruit Growing)	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Gardener and Nurseryman

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of the start of the 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of third semester.

** Minimum Credits required to get Certificate Course in Horticulture Sciences = 48 + 4 #.

Exit 1: Certificate Students can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during the first and second semester.

. ESSENTIAL COMPONENTS OF COURSE OF B.VOC

B.Voc. (Horticulture Sciences)
(SEMESTERS I &II)

PROGRAMME OUTCOMES

PO1. To transfer knowledge of Agriculture/Horticulture in the field of agricultural research, especially in Horticulture including Fruits, Vegetables, Flowers, Species, Medicinal & Aromatic Plants & their management.

PO2. To develop innovative agro-techniques to enhance the production & productivity of

Horticulture Crops. PO3. To increase farmer's income through Hi-tech Horticulture.

PO4.To create job opportunities for unemployed youth through teaching, research, training, extension, etc.

PO5. To the establishment of model nurseries in rural areas for the availability of quality planting materials.

PO6. To prolong the post-harvest storage life of Horticulture commodities & increase income through value addition of the products & to reduce post-harvest losses.

PO7. To develop interpersonal skills under the students to identify, analyze & solve proble

	COURSE TITLE	WEED CONTROL
1.	**Course Code**	HSC 101
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lecture) (Practical)**	30 hours of lectures for Theory + 60 hours of practical(1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks

(4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions.

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Gardener & Nursery Man.
7.	**Course Objectives**	To acquaint the students with elementary information on basic agronomic principles and practices for horticulture crops.
8.	**Course Outcome**	On successful completion of the course, the students will be able to: CO1. To understand about harmful & beneficial effects of weed in horticulture. CO2. To identify different weeds, cycle of weeds & crop competition. CO3. To understand different methods of weed control and Integrated Weed Management. CO4. To identify and apply herbicides for the control of weeds in different horticulture crops
9.	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT -1

Definition of weeds, their characteristics & harmful effects.
Classification of weeds, propagation & dissemination of weeds.

UNIT– II

Herbicide structure and properties, factors affecting the efficiency of herbicides, herbicide formulations, herbicide mixtures, Degradation of herbicides in soil and plants, weed shift in cropping systems, weed control through bioherbicides and allelochemicals, herbicide-resistant crops, herbicide rotation.

UNIT– III

Weed control/ management (Legislative, physical, chemical, mechanical, biological) in cropping systems and non–cropped situations–control of parasitic weeds – control of aquatic, biennial, and perennial weeds.

UNIT-IV

Weed control/ management in major horticultural crops. Integrated weed management

Practical work

1. Identification of important weeds and preparation of herbarium
2. Calculation of herbicide requirements
3. Use of various types of spray pumps and nozzles
4. Preparation of spray solutions and application of herbicides

Suggested Resources for Reading

1. Ashton, F.M and Crafts, A.S. 1981. Mode of Action of Herbicides (2nd Ed). Wiley Inter-Science
2. Gupta, O.P. 2007. Weed Management – Principles and Practices. Agro bios
3. Mandal, R. C, 1990. Weed, Weedicides and Weed Control – Principles and Practices. Agro–Botanical Publ.
4. Rao, V.S. 2000. Principles of Weed Science. Oxford & IBH

Note: Latest edition of textbooks may be used.

SEMESTER-I

	COURSE TITLE	NURSERY PRODUCTION
1.	**Course Code**	HSC 102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Self-Employment in the Business of Nursery.
7.	**Course Objectives**	To make the students familiar with the basic principles and procedures of nursery raising for horticulture crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Familiarization with principles & practices of propagation & nursery management for Horticultural Crops. CO2. Study of introduction to propagation, cellular basis for propagation, sexual propagation, apomixes, polyembryony, chimeras. CO3. Principal factors influencing seed germination of horticulture crops, dormancy, hormonal regulation of germination & seedling growth. CO4. Knowledge of nursery management, nursery establishment & nursery rules & regulations. CO5. Imparting knowledge about varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients and plant protection, hardening, packaging, storage, transport, and marketing of vegetables.

9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.
10.	**Course Content**	**UNIT– I** Scope and importance of vegetable and flower nursery planning I - location, soil, topography, layout, paths, roads, nursery beds, office, store, potting yard planning II - water, containers, electricity, fencing, shelter belts – Propagation structures - greenhouses, Mist chambers, poly houses – growing media - selection of media, quality of media, soil, sand, peat, leaf mold, saw dust, sphagnum moss, coco peat. **UNIT– II** Propagation – seed germination – dormancy-methods of breaking dormancy – pre-sowing seed treatment. methods of sowing – irrigation systems- nutrient management – water soluble and liquid fertilizers **UNIT– III** Varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients, packaging, storage, transport, and marketing of tomato, brinjal, chili, capsicum, cabbage, cauliflower, carrot, radish. **UNIT-IV** Varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients, packaging, storage, transport, and marketing of mango, apple, pear, peach, and guava. **Practical work:** 1. Features of nursery-factors to be considered for setting up of nursery 2. Studying the problems of nursery business 3. Visit to commercial vegetables, flower crops nursery 4. Seed sowing-vegetable crops-annual flower crops 5. Application of growth regulators for nursery 6. Irrigation and nutrient management-liquid fertilization 7. Care of nursery plants (shifting of plants, weeding, shade provision, watering, hardening). **Suggested Resources for Reading** 1. Nanda, K.K. and V.K. Kochhar; 1991. Vegetative propagation and management. Indus Publishing Company, New Delhi. 2. Sadhu, N.K. 1989. Plant propagation, Wiley Eastern Ltd., New Delhi. 3. Agarwal, R.L. 2005. Seed Technology. Oxford and IBH Publishing Co. New Delhi. **Note:** Latest edition of textbooks may be used.

	COURSE TITLE	IRRIGATION AND WATER MANAGEMENT
1.	**Course Code**	HSC 103
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practicals for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks, (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions.

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Technician in Irrigation Department.
7.	**Course Objectives**	To acquaint students with a soil-water relationship in the context to agriculture/ horticulture crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Impart knowledge and understanding of fundamental concepts and techniques of maintaining and enhancing water management. CO2. Develop the ability to skill the practices and their scope to improve in the water management. CO3.Impart knowledge and understanding of sources, modes, and systems of irrigation CO4. Familiarization with different methods of water conservation**.**
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT– I

Irrigation and its importance, Importance of water to plants, Time of irrigation. Principal of irrigation (soil moisture, root zone depth, water holding parameters, effective irrigation, daily water use) Basis for scheduling irrigation to crops: plant basis approach, climatologically approach and soil moisture depletion approach.

UNIT– II

Sources and modes of irrigation (canals, wells, tube wells, tank irrigation), irrigation system of Punjab, Flood irrigation, furrow irrigation, sub surface irrigation, overhead irrigation, manual irrigation, localized irrigation and sprinkler irrigation. Methods of measurement of irrigation, water requirement of different crops.

UNIT– III

Water logging, different types of water movement in soil. Principal of drainage: surface drainage and sub surface drainage. Kind of sub surface drains. Crop response to quality of irrigation water. Irrigation management and problem, Soil irrigation for frost protection.

UNIT-IV

Erosion Problems in Punjab and India, type of Erosion: water Erosion, Sheet and Rill Erosion, Scalding Erosion, Gully erosion, Tunnel Erosion, Stream Bank Erosion, Mass Erosion and Wind Erosion, Causes and Effects of Erosion, Factors responsible for Water and Wind Erosion. Form of wind and water Erosion. Different methods of water conservation.

Practical work:

1. Determination of soil moisture by different methods. Need for irrigation.
2. Measurement of irrigation water.
3. Preparation of land for different methods of irrigation.
4. Visit to area affected by erosion.
5. Visit to soil conservation research and demonstration center.

Suggested Resources for Reading

1. Iraeslon, Irrigation Principles, John Willey & Sons New Delhi 1996.
2. Gandhi R T., Gupta, P.C., Joseph, A.P. and Rage, N. I, Handbook of Irrigation WaterManagement.
3. Mickael, A.M., Irrigation Theory and Practices.

Note: Latest edition of text books may be used.

Semester II

<table>
<tr><th></th><th>COURSE TITLE</th><th>PROPAGATION TECHNIQUES</th></tr>
<tr><td>1.</td><td>Course Code</td><td>HSC-201</td></tr>
<tr><td>2.</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3.</td><td>Course Duration/ LP (Lectures) hrs./week)(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 2
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4.</td><td>Credits</td><td>4 Credits [2+2]</td></tr>
<tr><td>5.</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4x 2 Marks) = 08 Marks, (4x 8 Marks) = 32 Marks
i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.
Instructions for paper setters and candidates:
i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
</td></tr>
<tr><td>6.</td><td>Job Role</td><td>Gardeners</td></tr>
<tr><td>7.</td><td>Course Objectives</td><td>To get basic knowledge of various propagation techniques to develop practical skills.</td></tr>
<tr><td>8.</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:
CO1. Familiarization with principals & practices of propagation & nursery management for Horticultural Crops.
CO2. Study of introduction to propagation, cellular basis for propagation, sexual propagation, apomixes, polyembryony, chimeras.
CO3. Principal factors influencing seed germination of horticulture crops, dormancy, hormonal regulation of germination & seedling growth.</td></tr>
<tr><td>9.</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

10. Course Content

UNIT-1

Propagation: Need and Potentialities, Sexual and Asexual methods of propagation and their advantages and disadvantages Seed Dormancy: Causes (Internal and external factors) Seed Dormancy: Methods of breaking Dormancy

UNIT– II

Cutting Techniques: Stem Cutting, Leaf Cutting, root cutting Layering Techniques: tip layering, simple layering, compound layering, mound layering, air layering

UNIT– III

Grafting techniques, necessity of Grafting in plants and backstop. Type of Grafting: cleft Grafting, Bark Grafting, Side-veneer Grafting, Splice Grafting, Whip and Tongue Grafting, Saddle Grafting, Bridge Grafting.

UNIT-IV

Methods of Vegetative Propagation: Budding techniques: T or Shield Budding, Chip Budding, Inverted T-Budding, I-Budding, Flute Budding, Patch Budding. Advantages of Budding and safeguard during budding.

Practical work:

1. Horticultural tools and implements used for various operations.
2. Preparation of pot mixture, potting and repotting.
3. Preparation of growth regulators and method of application in horticultural crops.
4. Demonstration of propagation through layering and cutting.
5. Demonstration of propagation through budding.
6. Demonstration of propagation through grafting and top working.
7. Propagation through specialized plant parts.
8. Methods of manuring and irrigation.

Suggested Resources for Reading

1. Bose, T.K. Sanyal, D and Sandhu, M.L. (1998) Propagation of Horticultural crops. Naya Prakash Publishers, Kolkata
2. Sadhu, N.K. 1989. Plant propagation, Wiley Eastern Ltd., New Delhi.
3. Nanda K. K., Kochhar V. K. (1985). Vegetative Propagation of Plants. Kalyani Publisher- New Delhi- Ludhiana.
4. Hartmann, H.T., Kester D.E., Davis, F.T and R.L Geneve (2010). Plant Propagation: Principles and Practices (8th Edition)
5. Grewal, H.S. 1999. Propagation of ornamental plants, Kalyani Publishers, New Delhi
6. Hartmann, H.T. and D.E. Kester; 1989. Plant propagation-Principles and practices. Prentice – Hallof India, New Delhi.
7. Nanda, K.K. and V.K. Kochhar; 1991. Vegetative propagation and management. Indus Publishing Company, New Delhi.

Note: Latest edition of text books may be used.

COURSE TITLE	: ELEMENTS OF VEGETABLE GROWING
1. Course Code	HSC-202
2. Typology of Course	Core Discipline
3. Course Duration/ LP (Lectures) (Practical)	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4. Credits	4 Credits [2+2]
5. Max. Marks, Time and Instructions	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks, (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions.

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

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ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6. Job Role	Supervisor in Vegetable Industry.
7. Course Objectives	To get cultivations and other agronomic practices for raising vegetable crops.
8. Course Outcomes	On successful completion of the course, the students will be able to: CO1. To educate production technology of seasonal vegetables. CO2. To imparting knowledge about botany & taxonomy, climate & soil requirements, commercial varieties/hybrids, sowing/planting time & methods, seed rate & seed treatment, nutritional & irrigation requirements, intercultural operations, weed control, mulching, physiological disorders, harvesting, post harvest management, plant protection measures, & seed production of vegetables.
9. Pedagogy of the Course Work	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

10. Course Content

UNIT– I

Scope, Importance and Types of vegetable growing, importance of kitchen garden. Classification of vegetable crops.

UNIT– II

Cultivation practices including Soil and Climatic requirements, varieties, seed rate, time of sowing, spacing, fertilizers and manures, irrigation, hoeing and yield potential of Cole Crops, Root Crops, Bulb Crops and their harvesting.

UNIT– III

Cultivation practices including Soil and Climatic requirements varieties, seed rate, time of sowing, spacing,fertilizers and manures, irrigation, hoeing, and yield potential of Peas, Okra, eggplant and their harvesting.

UNIT-IV

Storage and seed production of the Potato, Cole Crops, Root Crops, Bulb Crops Peas and Solanaceae fruit

Practical work:

1. Identification of various vegetable crops, varieties and seeds.
2. Preparation of the field for sowing/ transplanting of various crops.
3. Application of manure and fertilizers, watering, hoeing, other cultural operations.
4. Prepare calendar of operation for various vegetable growing in Punjab.

Suggested Resources for Reading

1. Dhaliwal M.S., Handbook of vegetable crops, Kalyani Publishers, Ludhiana, 2008.
2. Package of Practices for Vegetable crops, P.A.U. Publications Ludhiana, Corresponding year.
3. Das, P.C., Vegetable crops of India, Kalyani Publishers, Ludhiana, 1993.
4. Chauhan, D.V., Vegetable production in India, S. Ram Prasad and Sons, Agra, 1993

Note: Latest edition of textbooks may be used.

<table>
<tr><th>COURSE TITLE</th><th>HORTICULTURE (FRUIT GROWING)</th></tr>
<tr><td>1. Course Code</td><td>HSC-203</td></tr>
<tr><td>2. Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3. Course Duration/ LP (Lectures) hrs./week)(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 2
(1 credit = 30 hours of practical based on 4 hrs./week) (1
Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4. Credits</td><td>4 Credits [2+2]</td></tr>
<tr><td>5. Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4x 2 Marks) = 08 Marks
(4x 8 Marks) = 32 Marks
i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions.
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

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ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
</td></tr>
<tr><td>6. Job Role</td><td>Horticulture Laboratories & Field Technician.</td></tr>
<tr><td>7. Course Objectives</td><td>To learn agronomic and other management practices in fruit growing. The
Student will also learn fruit preservation method & their scope.</td></tr>
<tr><td>8. Course Outcomes</td><td>On successful completion of the course, the students will be able to:
CO1. To understand the principle, and theoretical aspects & developing skills in fruit cultivation or horticultural crops.
CO2.To imparts basic knowledge about the importance & management of fruits grown in India..
CO3. To execute quality improvement by management practices.</td></tr>
<tr><td>9. Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
</table>

10 Course Content

UNIT– I

Fruits and their importance, Selection of site and soils for planting an orchard. Orchard soil management: clear culture, mulching, sod, sod mulch, intercropping, mixed cropping, multitier cropping, organic farming.

UNIT– II

Importance and scope of fruit growing in Punjab such as increase production, increase irrigation facilities, increase area under dry land, increase urbanization, increase transport facilities, increase cold storage facilities and pre cooling center, development new techniques, availability of loan, government incentive, export of fruits.

UNIT– III

History and development of fruit preservation and its scope. Canning of fruit (Gooseberry, Blueberry, Blackberry, Apple, Pear) Bottling of fruits (Banana jam, Lemon curd, Spiced citrus fruit, pickled Pears, Blueberry jam).

UNIT-IV

Different methods (canning, cooling, pasteurization, heating, boiling and adding sugar, salt and additives) of preservation and preparing jams, jellies and marmalades.

Practical work:

1. Identification of various vegetable crops, varieties and seeds. Preparation of the field for sowing/ transplanting of various crops.
2. Application of manure and fertilizers, watering, hoeing, other cultural operations. Prepare calendar of operation for various vegetable growing in Punjab.
3. Practice in propagation, methods, layout of orchard, transplanting, manuring, training and pruning of fruit plants.
4. Studies of cold storage and quick-freezing practices and refrigerated transport methods.

Suggested Resources for Reading

1. Singh, R., Fruits, National Book Trust of India, New Delhi, 1969.
2. Package of Practices for Cultivation of Fruits, P.A.U. Publications Ludhiana, Corresponding year.
3. Hayes, W.B., Fruit growing in India, Kitabistan, Allahabad, 1945.
4. Lal, G., Siddappa, S. And Tandon, G.L., Preservation of fruits and vegetables, Indian Council of Agricultural Research, New Delhi, 2009.

Note: Latest edition of text books may be used.

SEMESTER-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 301	Core Discipline	Insects & Pests of Horticultural & Vegetable Crops	2	2	4	40	10	40	10	100
HSC 302	Core Discipline	Landscaping & Floriculture	2	2	4	40	10	40	10	100
HSC 303	Core Discipline	Pomology-I	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC – III	Skill Enhancement Course	Common Basket			3					75
IDC III	Interdisciplinary Course	Common Basket			3					75
VAC – II	Value-Added Course	Common Basket			2					50
PW II	Project Work	Project Work-II& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

SEMESTER-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 401	Core Discipline	Introductory Soil Sciences	2	2	4	40	10	40	10	100
HSC 402	Core Discipline	Introductory Seed Technology	2	2	4	40	10	40	10	100
HSC 403	Core Discipline	Pomology-II	2	2	4	40	10	40	10	100
HSC 404	Core Discipline	Diseases and Management of Horticulture crops	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English/MIL			2					50
VAC – III	Value-Added Course	Common Basket			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Landscape & Floriculture Supervisor

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of the start of the 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of the 3rd semester.

**Minimum Credits required to get Diploma Course in Horticulture Sciences = 96 + 4 #.

#Exit 2: Diploma Students can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

COURSE TITLE		INSECTS AND PESTS OF HORTICULTURAL AND VEGETABLE CROPS
1.	**Course Code**	HSC 301
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> **Theory Paper:** **(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks** i) The exam of this course will be of 3 hours duration. ii) There will be no objective-type questions iii) Students are required to know about the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees. **Instructions for paper setters and candidates:** i) The syllabus of this paper has been divided into FOUR units. ii) All questions carry equal marks. iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each. iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
6.	**Job Role**	Supervisor in plant protection clinics
7.	**Course Objectives**	To make the students familiar with various insects -pests of horticulture and vegetables crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1: Familiarized with identification of different insect pests of horticulture, ornamentals, vegetables, and stored grains at the field level. CO2: To be able to understand the biology, diversity, distribution of insects, and their relationships to crops and the environmental conditions of a particular area. CO3: To understand identification of the nature of damage and symptoms caused by the pest so suitable technique of pest management can be applied for effective control. CO 4: Management of crop pests through the Integrated Pest Management
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT– I

Pest: definition and its categories, losses from pests to agricultural crops and their products. Natural Control and factors causing outbreak of pests of agricultural importance**.**

UNIT– II

Principle and methods of pest control in reference to IPM-its components viz. physical, legal, cultural, biological and chemical. Concept of economic injury and economic thresh hold level.

UNIT– III

Distribution, host range, damage and control of insect-pest of temperate fruits (apple root borer, apple stem bore; peach stem borer, peach fruit fly, cherry stem borer)

Distribution, host range, damage and control of insect pest of sub tropical fruits (citrus caterpillar, citrus psylla, litchi bug, bark eating caterpillar of loquat.)

UNIT-IV

Distribution, host range, damage and control of insect-pests of winter vegetables (potato tuber moth, potato aphid; onion maggots; pea leaf minor, turmeric leaf roller)

Distribution, host range, damage and control of insect-pests of summer vegetables (melon fruit fly, red pumpkin beetle: sweet potato weevil, chillies thrips, aphids)

Practical work:

1) Identification of various above-mentioned insect-pests, damaging stages, and their control measures
2) Preparation of an insect-pest box of various above-mentioned insect-pests

ESSENTIAL READINGS

1) Singh, R. Fruits, National Book Trust of India, New Delhi, 1969
2) Das, P.C., Vegetable crops of India, Kalyani Publishers, Ludhiana,1993
3) Package of Practices for Cultivation of Vegetable Crops, P.A.U. Publications Ludhiana, Corresponding year
4) Package of Practices for Cultivation of Fruits Crops, P.A.U. Publications Ludhiana, Corresponding year

SUGGESTED READINGS

1) Srivastva, K.P., Text book of Applied Entomology, Kalyani Publishers, Ludhiana, 2009
2) Atwal, A.S., and Dhaliwal, G.S., Agricultural pest of south Asia and their management, Kalyani Publishers Ludhiana, 2005

Note: Latest edition of textbooks may be used.

	COURSE TITLE	LANDSCAPING AND FLORICULTURE
1.	**Course Code**	HSC 302
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Landscaping and Floriculture Supervisor
7.	**Course Objectives**	To get basic principles in landscaping and knowledge for the cultivation of ornamental plants, shrubs and other floriculture plants.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1 The student will know fundamental gardening principles, which form the basis for learning landscape architecture. CO2 The students will be able to identify different soft and hard landscape elements. CO3 The students will become capable of designing a garden project and preparing the cost economics. CO The students prepare garden layouts to a specific scale for big commercial projects
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT– I

Principles of landscaping and floriculture, designs and styles of gardening, layout of home-gardens and indoor-scaping, layout of different styles of formal and informal gardens

UNIT– II

Utilization of important Trees, Shrubs, Climbers, Ground covers, Potted and Shade loving plants for gardening and landscaping

UNIT– III

Lawn management, selection of grass, soil preparation, planting of grass, diseases and other problems Importance of lawn in urban and rural areas and effect on trade

UNIT-IV

Commercial floriculture cultivation practices and propagation method of Chrysanthemum, Gladioli, Rose and Marigold each including detailed study of climate, soil, timing production, land preparation, different varieties and fertilization

Practical work:

1. Identification and maintenance of seasonal flowers, trees, shrubs, creepers & indoor plants
2. Raising of seedlings
3. Identification of implements and tools of landscaping and floriculture
4. Preparation of herbarium
5. Layout of home-gardens and lawns

ESSENTIAL READINGS

1. Salaria, A.S. 1999. Horticulure at Glance.M/s Jain Brothers, New Delhi
2. Des Raj 2002. Floriculture and Landscaping. 1st Edition, Kalyani Publishers, Ludhiana, India
3. Des Raj 2006. Objectives Floriculture and Landscaping, 1st Edition, Kalyani Publishers, Ludhiana,
4. India

.**SUGGESTED READINGS**

1. Arora, J.S. 1998. Introductory Ornamental Horticulture, Kalyani Publishers, Ludhiana
2. Gopalaswami, I. 2009. Complete Gardening in India - ICAR New Delhi
3. Randhawa, M.S. 2008. Beautiful trees and gardens, National book trust-New Delhi

Note: Latest edition of textbooks may be used.

	COURSE TITLE	POMOLOGY-I
1.	**Course Code**	HSC 303
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

The exam of this course will be of 3 hours duration.

1.There will be no objective-type questions

2. Students are required to know about the developments in the subject up to 6 months before the examination.

3. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

1.The syllabus of this paper has been divided into FOUR units.

2.All questions carry equal marks.

3. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

4. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Horticulture Inspector
7.	**Course Objectives**	To make the students familiar with basics of fruit production.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Impart basic knowledge about the importance and management of temperate fruits grown in India. CO 2. Study of commercial varieties of regional, national and international importance, ecophysiological requirements, recent trends in propagation, CO 3. Knowledge of industrial and export potential, Agri Export Zone
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10 Course Content

UNIT– I

Scope and importance of horticultural crops. Classification of horticultural plants based on botanical, geographical position and parts used. Horticultural zones of India and Punjab including hilly and high rainfall zone crops.

UNIT– II

Growth and development of horticultural plants – different stages of growth – juvenile phase, flowering and fruiting

Physiological changes in growth and development – respiration and photosynthesis. Factors influencing seed, dormancy and germination.

UNIT– III

Propagation –definition, merits and demerits – sexual and a sexual propagation. Seed propagation – seed treatments, sowing and seedling establishment. Vegetative propagation and factors influencing the success of vegetative propagation. Methods of vegetative propagation – cutting and layering. Methods of vegetative propagation – grafting and budding..

UNIT-IV

Micro propagation – merits, demerits and techniques

Cropping systems – intercropping, cover crop, multi-tier cropping practiced in horticultural crops.

Practical work:

1. Visit to Orchard and study of different components.
2. Horticultural tools and implements used for various operations.
3. Preparation of pot mixture, potting and repotting.

ESSENTIAL READINGS

1. Adams, C.R. and M. P. Early. 2004. Principles of Horticulture. Butterworth – Heinemam, Oxford University Press.
2. Chadha, K.L. 2001, Handbook of Horticulture, ICAR, New Delhi.
3. Chattopadhyaya, P.K.2001. A Text Book on Pomology (Fundamentals of Fruit Growing) Kalyani Publication, New Delhi
4. Christopher, E.P. 2001. Introductory Horticulture, Biotech Books, New Delhi
5. Edmond, J.B. T.L. Senn, F.S. Andrews and P.G.Halfacre, 1975. Fundamentals of Horticulture, Tata MC. Graw Hill Publishing Co.New Delhi

SUGGESTED READINGS

1. George Acquaah, 2002, Horticulture-principles and practices. Prentice-Half of India pvt. Ltd., New Delhi.
2. Hartman, H.T. and Kester, D.E. 1986. Plant Propagation – Principles and Practices – Prentice Hall of India Ltd., New Delhi.
3. Jitendra Singh. 2006. Basic Horticulture. Kalyani Publishers, New Delhi.
4. Rajan, S. and B.L. Markose. 2007. Propagation of Horticultural Crops. New India Publishing, New Delhi.

Note: Latest edition of textbooks may be used.

SEMESTER IV

	COURSE TITLE	INTRODUCTORY SOIL SCIENCE
1.	**Course Code**	HSC 401
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

1. The exam of this course will be of 3 hours duration.
2. There will be no objective-type questions
3. Students are required to know about the developments in the subject up to 6 months before the examination.
4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

1. The syllabus of this paper has been divided into FOUR units.
2. All questions carry equal marks.
3. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
4. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Soil Testing Technicians in laboratory
7.	**Course Objectives**	To teach introductory knowledge of weathering and study of properties of soil.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1: To be able about physical and chemical properties of soil and their effect on plant's health. CO2: To aware the students about causes, effects and remedies to prevention and mitigation of soil pollution. CO3: Knowledge about soil forming rocks and minerals, their weathering and soil forming processes and climatic factors affect them.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10 Course Content

UNIT– I

Definition & components of soil. Definition, occurrence & classification of soil forming rocks Soil Profile & various Soil Horizons

UNIT– II

Soil texture, soil structure, densities and porosity and factors affecting Importance of Soil structure, Soil Organic Matter origin, its nature and composition

UNIT– III

Decomposition of various organic compounds and Factors affecting decomposition, decompositions of water soluble and water insoluble organic matters, Advantages and disadvantages of organic matter

UNIT-IV

Plant growth concept and factors affecting plant growth. Plant nutrition: Essential elements, classification & general function of plant nutrients, forms of nutrient elements absorbed by plants, Process of nutrient uptake Soils of India.

Practical work:

1. Examination of soil profile in the field
2. Soil sampling and use of augers
3. Study of soil texture and soil structure of a given sample
4. Calculations of soil densities and porosity of soil
5. Soil survey of a farm

ESSENTIAL READINGS

1. Brady, N.C. and Weil, R.R., The Nature and Properties of Soil: 13th edn., Pearson education Pte. Ltd. New Delhi, 2002
2. Oswal, M.C., Soil Physics, Oxford & IBH publishing Co. Pvt. Ltd. New Delhi, 1994

SUGGESTED READINGS

1. Biswas, T.D., and Mukherjee, S.K., Text book of soil science, Tata McGraw Hill Publishing Co. Ltd., New Delhi,1997

Note: Latest edition of textbooks may be used.

	COURSE TITLE	INTRODUCTORY SEED TECHNOLOGY
1.	**Course Code**	HSC 402
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Supervisor in seed industries
7.	**Course Objectives**	To learn basic principles in seed production industry and release of varieties, their multiplications
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1: Start a seed production program for fill full the requirement of quality seed in market and increase the income. CO2: Storage the pure variety seed to avoid the availability crises of pure variety seed due to adverse environmental conditions. CO3: To supply disease-free seed in the market to promote the environmentally friendly cultivation of crops. CO4: To increase the farm income by producing high-yielding, disease-free quality seed and decreasing the cost of cultivation.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10 Course Content

UNIT– I

Definition, uses, morphology and types of seed
Seed health, factors affecting seed health, production of healthy seed Seed viability and germination of seed, factor affecting germination Seed vigour and factor affecting seed vigour

UNIT– II

Seed dormancy, Seed deterioration, its causes and prevention of seed deterioration Seed purity and purity test components

UNIT– III

Stages of production of improved seeds, Seed production organizations in India, Seed processing and seed treatments ,Packaging of seeds, Seed storage and factors affecting seed longevity in storage

UNIT-IV

Multiplication of seed (breeder seed, foundation seed, registered seed and certified seed) Seed certification and its objectives

Practical work:

1. Identification of different type of seeds
2. Study of method of seed health tests: - inspection of dry seeds, washing test and Blotter method
3. Seed purity test
4. Study of Seed germination test
5. Analysis Seed viability test.

ESSENTIAL READINGS

1. Vishunavat, K., Seed health testing, principal and protocol, Kalyani Publishers, New Delhi,2009
2. Sen Subir, and Ghosh N, Seed Science and technology, Kalyani Publishers, New Delhi, 2009

SUGGESTED READINGS

1. Singh, B.D., Plant Breeding Principles and Methods, Kalyani Publishers, New Delhi, 2009

Note: Latest edition of textbooks may be used.

	COURSE TITLE	POMOLOGY-II
1.	**Course Code**	HSC 403
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Horticulture Supervisor
7.	**Course Objectives**	To make the students familiar with basics of fruit production, green house management and post harvest handling in horticulture crops
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Impart basic knowledge about the importance and management of temperate fruits grown in India. CO 2. Study of commercial varieties of regional, national, and international importance, eco-physiological requirements, recent trends in propagation, CO 3. Knowledge of the production of bio-agents, bio-fertilizer, and greenhouse management, along with post-harvest handling in horticulture crops
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10 Course Content

UNIT– I

Principles of breeding of horticultural crops improvement of fruits. Importance of fruits and vegetable in human nutrition.

UNIT– II

Growth and Development physiology of flowering, fruit set and development, parthenocarpy and seedlessness.Maturity and ripening plant growth regulators and their role.

UNIT– III

Protected cultivation training, pruning and canopy management. Biotechnological tools for breeding for fruits. Micropropogation, Meristem culture, ovule culture, in vitro pollination.

UNIT-IV

Production of' bio-agents and bio-fertilizer and green house management. Importance of post harvest handling in horticulture crops.

Practical work:

1. Visit to identify temperate, tropical and sub tropical fruit plants.
2. Visit to fruit processing units.
3. Pruning practices in horticultural crops.
4. Maturity indices for various horticultural crops, handling, storage and packing techniques.
5. Visit to private orchards to identify different layout and cultural practices.

ESSENTIAL READINGS

1. Adams, C.R. and M. P. Early. 2004. Principles of Horticulture. Butterworth – Heinemam, Oxford University Press.
2. Chadha, K.L. 2001, Handbook of Horticulture, ICAR, New Delhi.
3. Chattopadhyaya, P.K.2001. A Text Book on Pomology (Fundamentals of Fruit Growing) Kalyani Publication, New Delhi
4. Christopher, E.P. 2001. Introductory Horticulture, Biotech Books, New Delhi
5. Edmond, J.B. T.L. Senn, F.S. Andrews and P. G. Halfacre, 1975. Fundamentals of Horticulture, Tata MC. Graw Hill Publishing Co.New Delhi

SUGGESTED READINGS

1. George Acquaah, 2002, Horticulture-principles and practices. Prentice-Half of India pvt. Ltd., New Delhi.
2. Hartman, H.T. and Kester, D.E. 1986. Plant Propagation – Principles and Practices – Prentice Hall of India Ltd., New Delhi.
3. Jitendra Singh. 2006. Basic Horticulture. Kalyani Publishers, New Delhi.
4. Rajan, S. and B.L. Markose. 2007. Propagation of Horticultural Crops. New India Publishing, New Delhi.
5. Singh, N.P. 2005. Basic Concepts of Fruit Science. International Book Distributing Co., Lucknow

Note: Latest edition of textbooks may be used.

	COURSE TITLE	DISEASES AND MANAGEMENT OF HORTICULTURE CROPS
1.	**Course Code**	HSC 404
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Supervisor in plant protection clinics
7.	**Course Objectives**	To acquaint with diseases of horticulture crops and their management
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1.Student will know the common pathogens of different diseases. CO 2. Student acquire the knowledge about etiology, and symptoms of these diseases which helps in diagnosis of the diseases of horticultural crops CO3. By knowing means of dispersal of these diseases suitable management methods can be applied. CO4. Eco-friendly and economically suitable management practices may be adopted
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10 Course Content

UNIT– I

Symptoms, host range their management and control of different diseases of fruits like apple, mango, citrus, papaya, sapota, grapes and guava.

UNIT– II

Symptoms, host range their management and control of different diseases of fruits like pear, peach, plum, apricot, ber, and loquat.

UNIT– III

Symptoms, host range their management and control of different diseases of vegetables like brinjal, melon, pumpkin, sweet potato, cabbage, potato, onion and peas.

UNIT-IV

Symptoms, host range their management and control of different diseases of vegetables like chilies and Turmeric.

Practical work:

1 Detailed study of symptoms and their management of horticultural and vegetables diseases of the above mentioned crops

2 Collection and dry preservation of diseased specimens of important crops.

ESSENTIAL READINGS

1. Gupta VK &.Sharma SK. 2000. Diseases of Fruit Crops. Kalyani Publ., New Delhi
2. Pathak VN. 1980. Diseases of Fruit Crops. Oxford & IBH, New Delhi
3. Singh RS. 2000. Diseases of Fruit Crops. Oxford & IBH, New Delhi

SUGGESTED READINGS

1. Walker JC. 2004. Diseases of Vegetable Crops. TTPP, India
2. Package of Practices for Vegetable Crops, P.A.U. Publications Ludhiana, Corresponding

Note: Latest edition of textbooks may be used.

PROPOSED SYLLABUS FOR B.VOC. (HORTICULTURE SCIENCE) -5TH & 6TH SEMESTER (NEP)

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 501	Core Discipline	Medicinal and Aromatic Plants	2	2	4	40	10	40	10	100
HSC 502	Core Discipline	Ornamental Horticulture	2	2	4	40	10	40	10	100
HSC 503	Core Discipline	Mushroom Cultivation Technology	2	2	4	40	10	40	10	100
HSC 504^	Core Discipline	Introductory Agroclimatology	2	2	4*	40	10	40	10	100
HSC 505^	Core Discipline	Commercial Floriculture	2	2	4*	40	10	40	10	100
HSC 506^	Core Discipline	Landscape Gardening	2	2	4*	40	10	40	10	100
VAC IV	Value-Added Course	Common Basket			2					50
PW III	Project Work	Project Work-III& Viva-Voce			2					50
Total Credits					**24**					**600**

^Any two in 5th Semester

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 601	Core Discipline	Horticulture Protected Cultivation	2	2	4	40	10	40	10	100
HSC 602	Core Discipline	Horticulture Post Harvest technology	2	2	4	40	10	40	10	100
HSC 603	Core Discipline	Recent Trends in Horticulture	2	2	4	40	10	40	10	100
HSC 604	Core Discipline	Organic Farming in Horticulture	2	2	4	40	10	40	10	100
HSC 605^	Core Discipline	Breeding of Horticultural Crops	2	2	4	40	10	40	10	100
HSC 606^	Core Discipline	Introductory Plant Biotechnology	2	2	4	40	10	40	10	100
HSC 607^	Core Discipline	Business Management in Horticulture	2	2	4	40	10	40	10	100
Total Credits					**24**					**600**

^Any two in 6th Semester

	COURSE TITLE	MEDICINAL AND AROMATIC PLANTS
1.	**Course Code**	HSC 501
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Self-Employment in the business of medicinal & aromatic nursery.
7.	**Course Objectives**	To acquaint students about different medicinal and aromatic crops, their package of practices
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Impart comprehensive knowledge about the production technology of medicinal and aromatic crops. CO2. Study of Herbal industry, Indian system of medicine, indigenous Traditional Knowledge, Classification of medicinal crops, Systems of cultivation CO3. Knowledge of production technology for medicinal and aromatic plants

9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.
10	**Course Content**	**UNIT– I** Scope and importance of medicinal and aromatic plants, classification of medicinal and aromatic plants according to botanical characteristics and uses, conservation of medicinal plants. **UNIT– II** Climate and soil requirements; agronomic practices; important chemical constituents and economic value of Aloe Vera (*Aloe barbadensis miller)*, Bahera (*Terminalia bellerica*), Stevia (*Stevia rebaudiana*), Giloy (*Tinospora cordifolia*), Black Musali (*Curculigo orchioides*), Thippali (*Piper longum. L.*) **UNIT– III** Climate and soil requirements; agronomic practices; important chemical constituents and economic value of rose (*Rosa spp.*), jasmine (*Jasminum spp.)*, chrysanthemum (*Chrysanthemum morifolium*), marigold (*Tagetes spp.***)**, tuberose (*Polianthes tuberosa***)** **UNIT-IV** Climate and soil requirements; agronomic practices; important chemical constituents and economic value of Celery (*Apium graveolens***)**, Coriander (*Coriandrum sativum* L.), Fennel (*Foeniculum vulgare)*, Funugreek (*Trigonella foenum- graecum L.*) **Practical work:** 1. Collection of locally available medicinal plants, Plant description and preparation of Herbarium. 2. Collection of locally available Aromatic plants, Plant description and preparation of Herbarium. 3. Study Agronomic practices of above mentioned medicinal and aromatic plants. 4. Study of propagation techniques of above-mentioned crops 5. Visit to some important and designed medicinal and aromatics gardens of Punjab and India. **ESSENTIAL READINGS** **1.** Chadha, K. L. and Gupta, R. ***Advances in Horticulture***. Vol. II. Medicinal and Aromatic Plants. Malhotra Publ. 1995. 2. Handa, S. S. ***Cultivation and Utilization of Medicinal Plants***. RRL, CSIR, Jammu. 1984. **SUGGESTED READINGS** 1. HussainA..Medicinal Plants and their Cultivation. CIMAP, Lucknow. 1993 2. Kumar, N., Khader, Md. A., Rangaswami, J.B.M. Irulappan. Introduction to Spices, Plantation Crops, Medicinal and Aromatic Plants. Oxford & IBH. 1997 3. Sharma, R. Agro-Techniques of Medicinal Plants. Daya Publ. House. 2004 **Note:** Latest edition of textbooks may be used.

COURSE TITLE		ORNAMENTAL HORTICULTURE
1.	**Course Code**	HSC 502
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Self-Employment in the Business of Ornamental Nursery.
7.	**Course Objectives**	To make the students familiar with the basic principles and procedures of ornamental gardening
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Acquire a critical knowledge of ornamental gardening and its significance. CO2. Identify and explain living and non-living components in an ornamental garden. CO3. Acquire skills on propagation and planting of various ornamental plants. CO4. Perform managerial skills related to ornamental gardening. CO5. Demonstrate skills of designing and developing ornamental gardens in public places.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT– I

Definition, scope and importance of gardening, aesthetic values; types of gardens in India. Landscaping, basic principles and basic components. Principles of gardening, garden components and adornments.

UNIT– II

Types of Ornamental gardens: Special types of gardens, trees, their design, their walk-paths, bridges, constructed features. Garden structures – greenhouse, glass house, net house.

UNIT– III

Plants in Ornamental gardens**:** Importance, design values, propagation, planting of following annuals, biennials and perennials:
(a) Climbers (b) Creepers (c) Palms (d) Ferns (e) Grasses (f) Cacti (g) Succulents

UNIT-IV

Ornamental gardening: Cultural operations in ornamental gardens and Beautifying educational institutions, near villages, urban planting railway stations, river banks, planting material for play grounds. Ornamental gardening in residences: Bottle garden, Vertical gardens, roof gardens, art of making bonsai.

Practical work:

1. Identification and description of various plants grown in ornamental gardens.
2. Tools, implements and containers used in ornamental gardening.
3. Demonstration of types and styles of gardens using photos or videos.
4. Preparation of land for lawn and planting.
5. Study and creation of bottle and vertical garden.

ESSENTIAL READINGS

1. Chadha, K.L. and Chaudhary, B. 1986. Ornamental Horticulture in India. Publication and Information division. ICAR, New Delhi.

2. K.V.Peter. 2009.Ornamental plants. New India Publishing Agency, New Delhi.

SUGGESTED READING

1. Arora, J.S. 2006. Introductory Ornamental Horticulture. Kalyani Publishers, Ludhiana

2. Bimaldas Chowdhury and Balai Lal Jana. 2014. Flowering Garden trees. Pointer publishers, Jaipur. India.

Note: Latest edition of textbooks may be used.

	COURSE TITLE	MUSHROOM CULTIVATION TECHNOLOGY
1.	**Course Code**	HSC 503
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Mushroom farm & processing / grower/ supervisor / technician /manager
7.	**Course Objectives**	To make the students familiar with the basic principles and procedures of nursery for horticulture crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Explain the biology, taxonomy and nutritional importance of edible mushroom CO2. Identify different types of edible and poisonous mushroom species CO3. Prepare substrates, compost and spawn preparations for various mushroom species. CO4. Analyze the economic, marketing and entrepreneurship opportunities in mushroom cultivation.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT I

Introduction to Mushroom- its history, scope and importance of mushroom cultivation in Punjab and India, the general biology, morphology and life cycles of mushroom (Oyster mushroom, Paddy straw mushroom, Button mushroom and Milky mushroom)

UNIT II

Cultivation technique of mushroom with special emphasis on Oyster mushroom, Paddy straw mushroom, button mushroom and Milky mushroom (including methods of preparing compost, spawn, spawning, cropping and harvesting)
Post-Harvest Technology- Preservation of Mushroom

UNIT III

Acquaintance with infrastructure, equipments and machineries required in the mushroom cultivation process
Nutritional value of mushrooms and the mushroom recipe

UNIT IV

Important Diseases, Pests, Mould of mushroom and their management
Growth of mushroom industry in Punjab and India

PRACTICAL

1. **1.** Training pertaining to entire process of Oyster mushroom, Paddy straw mushroom, Button mushroom and Milky mushroom cultivation.
2. **2.** One complete independent trial for mushroom cultivation
3. **3.** Visit to some important mushroom growing centres of Punjab and India

ESSENTIAL READINGS

1. Kaul, T.N..***Introduction to Mushroom Science (Systematics).*** Oxford & IBH Publishing Co. Pvt. Ltd. New Delhi & Calcutta, India. 1997
2. Vijaya Khader. ***Mushrooms for Livelihood.*** Kalyani Publishers, Ludhiana, India. 1998

SUGGESTED READINGS

1 Singh, Reeti and Singh,. ***Modern Mushroom Cultivation***. Agrobios, India. V.C. 2005

2 Suman, B.C. and Sharma, V.P. ***Mushroom Cultivation and Uses***. Agrobios, India. 2005.

Note: Latest edition of textbooks may be used.

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	COURSE TITLE	INTRODUCTORY AGROCLIMATOLOGY
1.	**Course Code**	HSC 504
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Agro climate assistant/ Farm advisory worker/Weather data recorder
7.	**Course Objectives**	To help students understand the relationship between climate and agriculture and apply basic weather knowledge for effective crop planning and management.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand the fundamental concepts of weather and climate, including temperature, rainfall, humidity, wind, and solar radiation and their influence on crop growth and development. CO2. Explain the relationship between climatic factors and different stages of crop production such as germination, flowering, and maturity. CO3. Interpret basic meteorological data and weather forecasts for making informed agricultural decisions. CO4. Analyze the effects of climate variability and climate change on agricultural productivity.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

11. Course Content

UNIT I

Importance of weather & climate, Different branches concerning atmospheric phenomenon. Division of climatology. Agro climatology: definition, scope and agro climatology as an interdisciplinary science. The earth and its atmosphere: structure & composition of earth. Motion of earth and their importance.

UNIT II

Atmospheric characteristics, physical properties and composition of atmosphere. Role of atmospheric air in agriculture. Elements of Climate and weather (solar radiation, air temperature, pressure, moisture & wind).Factors of climate. Effect of temperature, pressure, wind, moisture, solar radiation on crops.

UNIT III

Weather hazards (frost and high temperature, drought, floods, storms, tornadoes & water spouts, lighting, blizzard and earth-quakes).Effect of weather in agriculture industry.

UNIT IV

Climatic classification based on temperature, rain fall, vegetation. Climates of world with special reference to India. Agro climatic regions of Punjab and India. Role of weather and climate in sustainable agriculture.

PRACTICAL

1. Installation and use of various meteorological instruments for measurement of different weather parameters.
2. Study the principle and working of various meteorological instruments.
3. Study of wind direction and climatic water balance.
4. Meteorological satellite technology in weather forecasting.
5. Climatic maps of India and Punjab.

ESSENTIAL READINGS

1. Mavi, H.S.**,** ***Introduction to Agro meteorology,*** Oxford and IBH Publishing Co Pvt Ltd, New Delhi, 1994.
2. Critchfield, H. J., ***General Climatology***, Prentice Hall of India Pvt Ltd., New Delhi. 2002.

SUGGESTED READINGS

1. Mahi, G.S. and Kingra P.K, ***Fundamentals of Agro metrology,*** Kalyani Publisher, 2007.

Note: Latest edition of textbooks may be used.

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	COURSE TITLE	**COMMERCIAL FLORICULTURE**
1.	**Course Code**	HSC 505
2	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Floriculture Grower**,** Greenhouse/Polyhouse Manager**,** Floral Designer Quality Control Supervisor Marketing & Sales Executive**,** Post-Harvest Technologist
7.	**Course Objectives**	To impart knowledge and skills in scientific cultivation, post-harvest management, and marketing of commercial floriculture crops for profitable entrepreneurship.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Ability to grow flowers on a commercial scale. CO2.Understanding of nursery and greenhouse management. CO3.Knowledge of post-harvest handling and marketing of flowers. CO4. Capability to start and manage a floriculture enterprise.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT I

Scope and importance of commercial floriculture in India, production techniques of ornamental plants like rose, marigold, chrysanthemum, orchid, carnation, gladiolus for domestic and export market

UNIT II

Production techniques of ornamental plants like jasmine, dahlia, tuberose, bird of paradise, china aster and gerbera for domestic and export market.

UNIT III

Growing of flowers under protected environments such as glass house, plastic house, net house and green house

UNIT IV

Postharvest technology of cut flowers in respect of commercial flower crops, dehydration technique for drying of flowers, production techniques for bulbous.

PRACTICAL

1. Identification of commercially important floricultural crops.
2. Propagation practices in chrysanthemum, sowing of seeds and raising of seedlings of annuals.
3. Propagation by cutting, layering, budding and grafting. Training and pruning of roses.
4. Use of chemicals and other compounds for prolonging the vase life of cut flowers.
5. Drying and preservation of flowers.
6. Flower arrangement practices.

ESSENTIAL READINGS

1. Floriculture in India by G.S. Randhawa & A. Mukhopadhyay
2. Commercial Floriculture and Landscaping by Rajendra Singh
3. Advanced Commercial Floriculture (2 Volumes) by S.K. Bhattacharjee & L.C. De
4. Textbook on Commercial Floriculture and Ornamental Horticulture with Landscape Architecture by Ashok Kumar & Abhinav Kumar
5. Floriculture at a Glance by Desh Raj

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SUGGESTED READINGS

1. Protected Cultivation of Roses by T. Venkatesh Reddy & C.G. Nagaraja
2. Pests of Floriculture Crops and Their Control by Pulluri Ramesh
3. The Rose in India by B.P. Pal

Note: Latest edition of textbooks may be used.

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	COURSE TITLE	**LANDSCAPE GARDENING**
1.	**Course Code**	HSC 506
2	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Landscape gardener/ supervisor/ technician/ designer
7.	**Course Objectives**	To provide students with fundamental knowledge and practical skills in planning, designing, and maintaining sustainable landscape gardens.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand the principles and components of landscape gardening. CO2. Identify and select suitable ornamental plants for different landscape uses. CO3. Design basic landscape layouts for homes, institutions, and public spaces. CO4. Establish and maintain lawns, gardens, and landscape features. CO5. Apply sustainable and eco-friendly landscaping practice
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

11. Course Content

UNIT I

Principles and elements of landscape design, gardens elements and designs, propagation of ornamental plants, graphics and symbols in landscape gardening, tools and implements used in landscape design

UNIT II

Layout of formal gardens, informal gardens, special type of gardens, layout of rock garden and gravel garden

UNIT III

Layout of roof gardening , Paved garden, Marsh or Bog garden , landscaping of places of public importance

UNIT IV

Designing of greenhouse, Conservatory and lathe house , gardening of public places, Lawn: establishment and maintenance, CAD application

PRACTICAL

1. Identification of tools and implements used in landscape design,
2. Training and pruning of plants for special effects,
3. Lawn establishment and maintenance,
4. Layout of formal gardens, informal gardens, special type of gardens (sunken garden, terrace garden, rock garden) and designing of conservatory and lathe house.
5. Use of computer software, visit to important gardens/ parks/institutes.

ESSENTIAL READINGS

1.Bose, T. K., Maiti, R. G., Dhua, R. S. & Das, P. 1999. Floriculture and Landscaping. Naya Prakash.
2. Chadha, K. L. & Chaudhury, B. 1992. Ornamental Horticulture in India. ICAR.
3. Chadha, K. L. 1995. Advances in Horticulture. Vol. XII. Malhotra Publ. House.

SUGGESTED READINGS

1.Randhawa, G. S. & Mukhopadhyay, A. 1986. Floriculture in India. Allied Publ.
2. Reddy, S., Janakiram, B., Balaji, T., Kulkarni, S. & Misra, R.L. 2007. Hightech Floriculture. Indian Society of Ornamental Horticulture, New Delhi.
3. Nambisan, K.M.P. 1992. Design Elements of Landscape Gardening. Oxford &IBH.

Note: Latest edition of textbooks may be used.

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SEMESTER VI

	COURSE TITLE	**HORTICULTURE PROTECTED CULTIVATION**
1.	**Course Code**	HSC 601
2	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

1) The exam of this course will be of 3 hours duration.
2) There will be no objective-type questions
3) Students are required to know about the developments in the subject up to 6 months before the examination.
4) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

1) The syllabus of this paper has been divided into FOUR units.
2) All questions carry equal marks.
3) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
4) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Protected Cultivation Technician/Greenhouse Manager/ Nursery Supervisor
7.	**Course Objectives**	To develop knowledge and practical skills in protected cultivation techniques for efficient production of high-value horticultural crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Ability to understand principles and structures of protected cultivation. CO2. Skill in managing greenhouse climate, irrigation, and fertigation. CO3. Knowledge of pest, disease, and nutrient management under protected conditions. CO4.Capability to produce high-quality horticultural crops profitably.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10.Course Content

UNIT I

Green house technology: Introduction, Types of Green House, Plant response to Greenhouse environment, Planning and design of greenhouses

UNIT II

Green house equipment, materials of construction for traditional and low cost green houses, Irrigation systems used in greenhouses

UNIT III

Growing media, soil culture, type of soil required, drainage, flooding and leaching, soil pasteurization in peat moss and mixtures, rock wool and other inert media, nutrient film technique (NFT) / hydroponics

UNIT IV

Cost estimation and economic analysis. Choice of horticulture crops for cultivation under greenhouses, problems / constraints of greenhouse cultivation and future strategies.

PRACTICALS

1. Study of different types of greenhouses based on shape, construction and cladding materials
2. Testing of soil and water to study its suitability for growing crops in greenhouses
3. The study of fertigation requirements for greenhouses crops
4. The study of various growing media used in raising of greenhouse crops and their preparation and pasteurization / sterilization
5. Visit to commercial green houses

ESSENTIAL READINGS

1. Balraj Singh. ***Protected cultivation of vegetable crops***. Kalyani Publishers, Ludhiana. 2006
2. Brahma Singh. ***Advances in Protected Cultivation***. New India Publishing Agency. New Delhi. 2014
3. Reddy P. Parvatha. ***Protected Cultivation***. Springer Publications. USA. 2003.

SUGGESTED READINGS

1. Jitendra Singh, ***Precision Farming in Horticulture***. New India Publishing Agency. New Delhi. 2015.
2. T. Pradeep Kumar, B. Suma, Jyothi Bhaskar and K.N.Satheson. ***Management of Horticultural crops***. New India Publishing Agency, New Delhi. 2008

Note: Latest edition of textbooks may be used.

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	COURSE TITLE	**HORTICULTURE POST HARVEST TECHNOLOGY**
1.	**Course Code**	HSC 602
2	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.

ii) There will be no objective-type questions

iii) Students are required to know about the developments in the subject up to 6 months before the examination.

iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.

ii) All questions carry equal marks.

iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Post-Harvest Management Officer /Quality Control Supervisor/Cold Storage Technician/Packaging & Processing Assistant
7.	**Course Objectives**	To develop knowledge and skills in handling, storage, processing, and marketing of horticultural produce to reduce post-harvest losses and maintain quality.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Ability to understand principles of post-harvest handling of horticultural crops. CO2. Knowledge of grading, packaging, storage, and transportation methods. CO3.Skill in reducing post-harvest losses and maintaining produce quality. CO4. Capability to manage post-harvest units and marketing of produce.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10 Course Content

UNIT I

Importance of postharvest technology in horticultural crops. Maturity indices, harvesting, handling, grading of fruits, vegetables, cut flowers, plantation crops, spices, medicinal and aromatic plants

UNIT II

Pre-harvest factors affecting quality, factors responsible for deterioration of horticultural produce, physiological and biochemical changes, hardening and delaying ripening process.

UNIT III

Quality parameters and specifications. Structure of fruits, vegetables and cut flowers related to physiological changes after harvest. Methods of storage for local market and export.

UNIT IV

Pre-harvest treatment, pre-cooling, pre-storage treatments. Different systems of storage, packaging methods, and types of packages, recent advances in packaging.

PRACTICAL

1. Practice in judging the maturity of various horticultural produce, determination of physiological loss in weight and quality.
2. Grading of horticultural produce, post-harvest treatment of horticultural crops, physical and chemical methods.
3. Packaging studies in fruits, vegetables, plantation crops, spices and cut flowers by using different packaging materials, methods of storage, post-harvest disorders in horticultural produce.
4. Identification of storage pests and disease insects.

ESSENTIAL READINGS

1 Verma, L. R. and Joshi, V. K.Post Harvest Technology of Fruits and Vegetables. Vol. I & I. Indus Publishing Co., New Delhi 2000

2 Stanley, J. K. Post Harvest Physiology of Perishable Plant Products. CBS, New Delhi. 1998

3. Neetu Sharma and MashkoorAlam, M..Post Harvest Diseases of Horticultural 1998

SUGGESTED READINGS

1 Pruthi, J. S. Minor Spices and Condiments – Crop Managements and Post Harvest Technology. ICAR, New Delhi. 2001

2 Saraswathy, S. *et. al*.. Post harvest Management of Horticultural Crops. Agribios (India).81-7754-322-9. 2008

Note: Latest edition of textbooks may be used.

	COURSE TITLE	**RECENT TRENDS IN HORTICULTURE**
1.	**Course Code**	HSC 603
2	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks, Time and Instructions**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6.	**Job Role**	Horticulture technicians, farm managers, agri-entrepreneurs, extension workers, and professionals in protected cultivation and agri-based industries.
7.	**Course Objectives**	To familiarize learners with modern technologies, sustainable practices, post-harvest innovations, and emerging market trends shaping contemporary horticulture.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO 1. Understand recent technological innovations and modern production systems in horticulture. CO 2. Apply sustainable and climate-smart practices in horticultural crop management.
9.	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT I

Precision horticulture: concept and applications, GIS, GPS, and remote sensing in horticulture, Smart irrigation -fertigation systems, Soilless cultivation: hydroponics, aeroponics, aquaponics

UNIT II

Organic horticulture and certification standards, Sustainable soil and nutrient management, Biofertilizers, Protected cultivation: polyhouse, greenhouse, shade net, Water conservation techniques

UNIT III

Post-harvest of horticultural crops, Advanced storage and cold chain management, Packaging innovations, and biodegradable packaging

UNIT IV

Urban and vertical farming systems, Medicinal and aromatic plants sector, Entrepreneurship and startups in horticulture, Digital marketing and e-commerce of horticultural produce, Government policies and institutional support

PRACTICALS

1. Study of vertical farming structures, growing media, crop selection, lighting, irrigation, and nutrient management.
2. Observation and recording of growth parameters, yield, resource use efficiency, and comparison with conventional cultivation methods.

ESSENTIAL READINGS

1. **Current Trends in Horticulture —** Amit Kumar Bhargava *(covers modern practices and innovations)*
2. **Trends in Horticulture —** Edited by P. Sudheer Kumar Reddy et al *(comprehensive overview of recent developments)*
3. **Handbook of Horticulture —** K. L. Chadha *(ICAR comprehensive text on horticulture fundamentals and modern practices)*

SUGGESTED READINGS

1. **Ornamental Horticulture: Latest Cultivation Practices and Breeding Technologies** — Bhavya Bhargava et al *(recent advancements in floriculture & ornamentals)*
2. **Recent Trends in Vegetable Production** — S.P. Kanaujia, Raj Narayan & others *(focus on modern vegetable production technology)*
3. **Glaustas Horticulture** — P. Muthukumar & R. Selvakumar *(core horticulture topics with practical examples)*

Note: Latest edition of textbooks may be used.

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<table>
<tr><th colspan="2">COURSE TITLE</th><th>ORGANIC FARMING IN HORTICULTURE</th></tr>
<tr><td>1.</td><td>Course Code</td><td>HSC 604</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3.</td><td>Course Duration/ LP (Lectures) hrs./week) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for Practical for each unit</td></tr>
<tr><td>4.</td><td>Credits</td><td>4 Credits [2+2]</td></tr>
<tr><td>5.</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks
i) The exam of this course will be of 3 hours duration.
ii) There will be no objective-type questions
iii) Students are required to know about the developments in the subject up to 6 months before the examination.
iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.
Instructions for paper setters and candidates:
i) The syllabus of this paper has been divided into FOUR units.
ii) All questions carry equal marks.
iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
</td></tr>
<tr><td>6.</td><td>Job Role</td><td>Prepares learners for employment as organic farm managers, horticulture extension workers, quality assurance or certification assistants, organic input entrepreneurs, and sustainable agriculture practitioners.</td></tr>
<tr><td>7.</td><td>Course Objectives</td><td>To develop skills for sustainable production, certification, and marketing of organic horticultural crops.</td></tr>
<tr><td>8.</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to:

CO1. Understand concepts and standards of organic horticulture.

CO2.Apply organic practices for soil, nutrient, pest, and crop management.</td></tr>
<tr><td>9.</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.</td></tr>
</table>

10.Course Content

UNIT I

Concept, definition, and principles of organic farming; history and development of organic horticulture; scope and importance in India; organic vs conventional horticulture; organic standards and certification systems (NPOP, PGS-India)

UNIT II

Soil health and organic matter management; composting methods (FYM, vermicompost, NADEP); green manuring and cover crops; biofertilizers and microbial inoculants; organic nutrient sources and nutrient budgeting.

UNIT III

Principles of organic pest and disease management; cultural, mechanical, and biological methods; biopesticides and botanicals; beneficial insects and biocontrol agents; organic weed management practices.

UNIT IV

Organic production of fruits, vegetables, spices, and flowers; organic protected cultivation; harvesting and post-harvest handling; certification procedures and documentation; marketing, export potential, and organic value chains.

PRACTICALS

1 Preparation and application of organic inputs: Compost, vermicompost, NADEP compost, and liquid manures (Jeevamrit, Panchagavya).

2. **Identification and use of biofertilizers and biopesticides**: Study of Rhizobium, Azotobacter, PSB, Trichoderma, neem-based formulations, and botanical extracts.

3. Organic crop management practices: Seed treatment, organic nutrient scheduling, pest and disease management, and weed control in horticultural crops.

ESSENTIAL READINGS

1. **Principles of Organic Farming (Textbook)** – P. L. Maliwal *(ICAR syllabus-aligned core text)*
2. **Organic Farming: Principles and Practices** – A. Dinesh, M. Kamala Priya & S. Alekya Naidu *(comprehensive student-oriented guide)*
3. **Organic Horticulture: Principles, Practices and Technologies** – H. P. Singh & George V. Thomas *(focuses on organic horticultural systems)*

SUGGESTED READINGS

1. **Organic Farming: Sustainable Agriculture and Horticulture Production** – Rameshwar Dass Gupta *(covers organic farming principles linked to horticulture)*
2. **Sustainable Horticulture: A Key to Futuristic Farming** – Afiya R. S., Senthilkumar S. & Manivannan S *(includes chapters on organic inputs, certification and crop management)*

Note: Latest edition of textbooks may be used.

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COURSE TITLE	BREEDING OF HORTICULTURAL CROPS
1. Course Code	HSC 605
2 Typology of Course	Core Discipline
3. Course Duration/ LP (Lectures) hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4. Credits	4 Credits [2+2]
5. Max. Marks, Time and Instructions	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> **Theory Paper:** **(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks** i) The exam of this course will be of 3 hours duration. ii) There will be no objective-type questions iii) Students are required to know about the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees. **Instructions for paper setters and candidates:** i) The syllabus of this paper has been divided into FOUR units. ii) All questions carry equal marks. iii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each. iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
6. Job Role	Prepares students for careers as horticultural plant breeders, seed production and quality control professionals, research assistants, extension officers, and nursery or germplasm managers in public and private sectors.
7. Course Objectives	To provide fundamental knowledge of principles and methods of plant breeding as applied to horticultural crops.
8. **Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand genetic principles and breeding methods used in horticultural crops. CO2. Apply crop improvement strategies for yield, quality, resistance, and adaptability.
9. Pedagogy of the Course Work	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.

10. Course Content

UNIT I

Importance and objectives of breeding horticultural crops; reproductive biology and floral mechanisms; modes of reproduction and pollination; genetic variability and heritability; introduction and acclimatization.

UNIT II

Selection methods (mass, pure line, clonal selection); hybridization techniques; heterosis breeding; polyploidy and mutation breeding; breeding for resistance to biotic and abiotic stresses.

UNIT III

Breeding methods and achievements in major fruit crops (mango, citrus, banana); vegetable crops (tomato, brinjal, cucurbits); ornamental crops (rose, chrysanthemum); plantation and spice crops—objectives and improved varieties.

UNIT IV

Biotechnological approaches in breeding; tissue culture and micropropagation; marker-assisted selection; transgenic crops and genome editing (introductory); seed production, variety release procedures, and intellectual property rights (PPV&FR Act).

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PRACTICALS

1 Study of floral biology and breeding behaviour of horticultural crops, including flower structure, anthesis, pollination, and mode of reproduction.

2 **Demonstration of breeding techniques** such as selection methods, emasculation, controlled pollination, and hybrid seed production in horticultural crops.

ESSENTIAL READINGS

1. **Breeding of Horticultural Crops: Principles and Practices** — N. Kumar (2nd Revised & Expanded Ed) — comprehensive text on breeding methods and crop improvement in horticulture.
2. **Breeding of Horticultural Crops: Principles and Practices** — Bijendra Kumar Singh & Rajaneesh Singh — latest overview including reproduction, breeding systems, pest resistance, and seed production.
3. **Horticultural Plant Breeding** — Thomas J. Orton — detailed global perspective on breeding objectives, germplasm, population engineering and cultivar development.

SUGGESTED READINGS

1. **Plant Breeding in Horticulture** — H. Ozaktan, A. Akkopru & N. Ustun (International Society for Horticultural Science) — multi-author compilation on breeding programs and genomics/tools.
2. **Breeding of Horticultural Crops: Vol. 2 — Tropical Fruits** — Edited by V. A. Parthasarathy et al. — focus on tropical fruit breeding strategies and case studies.

Note: Latest edition of textbooks may be used.

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COURSE TITLE	INTRODUCTORY PLANT BIOTECHNOLOGY
1. Course Code	HSC 606
2 Typology of Course	Core Discipline
3. Course Duration/ LP (Lectures) hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4. Credits	4 Credits [2+2]
5. Max. Marks, Time and Instructions	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

Theory Paper:

(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks

v) The exam of this course will be of 3 hours duration.

vi) There will be no objective-type questions

vii) Students are required to know about the developments in the subject up to 6 months before the examination.

viii) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees.

Instructions for paper setters and candidates:

v) The syllabus of this paper has been divided into FOUR units.

vi) All questions carry equal marks.

vii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.

viii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

6. Job Role	Prepares students for entry-level roles as tissue culture technicians, nursery biotechnology assistants, quality control staff, and field-level horticulture professionals applying basic plant biotechnology tools
7. Course Objectives	To introduce basic concepts and tools of plant biotechnology relevant to horticulture.
8. **Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand fundamental principles and techniques of plant biotechnology. CO2. Apply biotechnological tools in horticultural crop production and improvement.
9. Pedagogy of the Course Work	Classroom Lecture, Laboratory, Interactive Sessions, PPT, Practical, and Group Discussions, Case Studies, etc.

10. Course Content

UNIT I

Definition, scope, and importance of plant biotechnology; history and milestones; basic concepts of cell, tissue, and molecular biology; plant cell structure; totipotency; ethics and biosafety in biotechnology.

UNIT II

Principles and types of plant tissue culture; laboratory requirements; culture media and growth regulators; sterilization techniques; micropropagation stages; somaclonal variation; applications of tissue culture in horticulture.

UNIT III

Basics of recombinant DNA technology; restriction enzymes, vectors, and cloning; gene transfer methods (Agrobacterium-mediated, biolistics); PCR and molecular markers (introductory); genetically modified plants—advantages and concerns.

UNIT IV

Biotechnology in crop improvement for yield, quality, and stress resistance; virus-free planting material; marker-assisted selection (introductory); biofertilizers and biopesticides; role of biotechnology in sustainable horticulture

PRACTICALS

1. Preparation of culture media and study of laboratory equipment
2. Demonstration of explant sterilization and micropropagation
3. Visit to tissue culture laboratory/nursery

ESSENTIAL READINGS

1. **Introduction to Plant Biotechnology** — H. S. Chawla (comprehensive undergraduate text on plant tissue culture, micropropagation, molecular techniques, and applications).
2. **Plant Biotechnology: Principles and Applications** — Springer (holistic coverage from basics to modern applications, including genetic engineering and biosafety).

SUGGESTED READINGS

1. **Biotechnological Advances in Horticultural Crops —** Edited by Ajay Kumar Thakur & Co. *(focus on horticultural crop applications and innovations)*.
2. **Plant Biotechnology in Ornamental Horticulture —** Yi Li & Yan Pei *(biotechnology applications specific to ornamental plants)*.

Note: Latest edition of textbooks may be used.

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COURSE TITLE	BUSINESS MANAGEMENT IN HORTICULTURE
1. Course Code	HSC 607
2 Typology of Course	Core Discipline
3. Course Duration/ LP (Lectures) hrs./week) **(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 2-groups will be created for Practical for each unit
4. Credits	4 Credits [2+2]
5. Max. Marks, Time and Instructions	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> **Theory Paper:** **(4x 2 Marks) = 08 Marks , (4x 8 Marks) = 32 Marks** i) The exam of this course will be of 3 hours duration. ii) There will be no objective-type questions iii) Students are required to know about the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculator will not be provided by the University/ College to the examinees. **Instructions for paper setters and candidates:** i) The syllabus of this paper has been divided into FOUR units. ii) All questions carry equal marks. iii) The examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. In Question No. 1, students are required to attempt 4 out of 6 short questions of 2 marks each. iv) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
6. Job Role	Prepares students for roles as horticulture business managers, agripreneurs, marketing executives, farm enterprise supervisors, and startup professionals in horticulture-based industries.
7. Course Objectives	To familiarize students with planning, marketing, finance, and management of horticulture-based businesses.
8. **Course Outcomes**	On successful completion of the course, the students will be able to: CO1 Understand principles of management and entrepreneurship in horticulture. CO2 Apply business management skills to start, manage, and sustain horticultural enterprises.
9. Pedagogy of the Course Work	Classroom Lecture, Laboratory, Interactive Sessions, PPT, Practical, and Group Discussions, Case Studies, etc.

10. Course Content

UNIT I

Concept and functions of management; types of business organizations; principles of management; role of management in horticulture enterprises; leadership, motivation, and communication skills.

UNIT II

Business planning and feasibility analysis; project formulation for horticulture enterprises; cost of cultivation and budgeting; sources of finance; basics of accounting, record keeping, and financial statements.

UNIT III

Concepts of agricultural and horticultural marketing; market structure and price determination; post-harvest handling and value addition; supply chain management; e-marketing and digital platforms for horticultural produce.

UNIT IV

Entrepreneurship concepts and startup ecosystem; agripreneurship in horticulture; risk management and insurance; government schemes, subsidies, and institutional support; legal aspects, licensing, and quality standards.

PRACTICALS

1. Preparation of a simple business plan for a horticulture enterprise
2 Case study of a successful horticulture entrepreneur/startup
3 Field visit to a horticulture-based enterprise or market

ESSENTIAL READINGS

1. **Fundamentals of Business Management —** S. C. Gupta & N. K. Gupta *(core management principles)*
2. **Agribusiness Management —** Anil Kumar, R. A. Sharma & others *(focus on agribusiness models and marketing)*
3. **Entrepreneurship Development and Management —** Vasant Desai *(entrepreneurial skills and business planning)*
4. **Agricultural and Horticultural Marketing in India —** S. S. Acharya & N. L. Agarwal *(market systems, pricing, and post-harvest marketing)*

SUGGESTED READINGS

1 Business Management and Entrepreneurship — Dr. C. B. Gupta *(management theory + entrepreneurship*
2 Rural Marketing: Environment, Problems and Strategies — C. G. Krishnamacharyulu & Lalitha Ramakrishnan *(rural & horticulture marketing focus)*
3 Management Accounting and Financial Control — M. Y. Khan & P. K. Jain *(financial decision-making tools)*

Note: Latest edition of textbooks may be used.

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